

Faculty Vitae: Vikas Berry

1. Education

BS Chemical Engineering, Indian Institute of Technology-Delhi, 1999
MS Chemical Engineering, University of Kansas, 2003
PhD Chemical Engineering, Virginia Tech, 2006

2. Academic Experience

University of Illinois at Chicago, Chicago, IL

- Aug 2022 – Present Dr. Satish C. & Asha Saxena Professor
- Nov 2016 – Present Department Head
- Aug 2019 – Present Professor
- Aug 2015 – Nov 2016 Interim Department Head
- Aug 2014 – Aug 2019 Associate Professor

Kansas State University, Manhattan, KS

- Nov 2011 – Aug 2014 William H. Honstead Professor
- Jul 2012 – Aug 2014 Associate Professor
- Jan 2007 – Jul 2012 Assistant Professor

Virginia Tech, Blacksburg, VA

- Jan 2013 – Dec 2016 Graduate Research Assistant (PhD)

University of Kansas, Lawrence, KS

- Aug 2000 – Dec 2002 Graduate Research Assistant (MS)

3. Non-Academic Experience

Grapherry Inc. Mar 2024 – Present. Co-Founder and Act. Chief Executive Officer

Viroscope Inc. Oct 2023 – Present Co-Founder

Cadila Pharmaceuticals, India Aug 1999 – Jul 2000 Pharmaceutical Process Scientist

4. Certification or professional registrations

None

5. Current membership in professional organizations

- AIChE, Electrochemical Society, Omega Chi Epsilon Honor Society of Chemical Engineering, Member, Sigma Xi Scientific Research Society, Member, 2007

6. Honors and Awards

- Listed in the Top 2% cited researchers since 1960 (Stanford Rankings), 2021
- Louis Stokes Alliances for Minority Participation-NSF International Center of Excellence (LSAMP-NICE) Faculty Advisor Award, 2019
- Rudolph A. Marcus Award, 2017
- NSF-CAREER Award, 2011-17
- William H. Honstead Professorship, 2011 – 2014
- Sigma Xi Outstanding Junior Scientist Award, 2010
- Big-XII Fellow - 2009, Kansas State University
- Stroebel Scholarship Recipient, University of Kansas 2000-02

7. Selected service activities

- 8 research group alumni currently hold faculty positions
- Editorial Service
 - Scientific Reports, Editorial Board Member, 2012 – Present
 - Science Advances Letters, Associate Editor, 2014 – Present
 - Advanced Carbon, Scientific Advisory Board
 - Journal of Nanomaterials and Molecular Nanotechnology, Scientific Advisory Board

8. Selected Publications and Presentations

SELECT PUBLICATIONS (CITATIONS = 9332) *Indicates corresponding authorship

1. Mehran Arzani, Hamidreza Mahdavi and Vikas Berry, “Engineering porous liquids for enhanced mobility and stable battery electrolytes”, **ACS Energy Letters**, 10, 3259, 2025
2. Hisham Maddah, Vikas Berry and Sanjay Behura, “Optimal design of spin-coated layers of TiO₂ for maximum open-circuit voltage in anthocyanin-based dye-sensitized solar cells”, **Energy Conversion and Management**, 346, 120478, 2025
3. Kartikey Sharma, Songwei Che, Sanjay Behura, Naveen Dandu, Anika Shinde, Anh Ngo and Vikas Berry, “h₆ Organometallic Functionalization of Hexagonal Boron Nitride: Implications for Multifunctional Molecular and Nanoscale Interfaces for Electronic Applications”, **ACS Applied Nano Materials**, 1, 233, 2024
4. Ahmad Jaradat, Musawenkosi K Ncube, Ilias Papailias, Nikhil Rai, Khagesh Kumar, Volodymyr Koverga, Roshan Y Nemade, Chengji Zhang, Nannan Shan, Hessam Shahbazi, Arash Namaeighasemi, Pardis Seraji, Shahriar Namvar, Vikas Berry, Jordi Cabana, Arunkumar Subramanian, Anh T Ngo, Larry A Curtiss, Amin Salehi-Khojin, “Fast Charge-Transfer Rates in Li-CO₂ Batteries with a Coupled Cation-Electron Transfer Process”, **Advanced Energy Materials**, 14 (15). 2303467, 2024

5. Roshan Nemade, Sheldon Cotts, and Vikas Berry, "Graphene Fermi Level-Guided Attachment of Single Exoelectrogens and Induced interfacial Doping", **ACS Applied Materials Interfaces**, 16 (5). 5548 2024
6. Sungjoon Kim, Vikas Berry, Jessica Metcalfe, and Anirudha Sumant, "Thin Film Charged Particle Detectors", *Journal of Instrumentation*, 18, 07, P07047, 2023
7. Nishithan C Kani, Ngoc Hoang Lan Nguyen, Kyle Markel, Rajan R Bhawnani, Benjamin Shindel, Kartikey Sharma, Sungjoon Kim, Vinayak P Dravid, Vikas Berry, Joseph A Gauthier, Meenesh R Singh, "Electrochemical Reduction of Nitrates on CoO Nanoclusters-Functionalized Graphene with Highest Mass Activity and Nearly 100% Selectivity to Ammonia", **Advanced Energy Materials**, 13, 17, 2204236
8. Hisham A Maddah, Vikas Berry, Sanjay K Behura, "Simple harmonic oscillation model explaining MA torsional locking in surface passivated MAPbI₃ crystal", **Chemical Physics Letters**, 806. 139967, 2022
9. Bijentimala Keisham, Sungjoon Kim, Sheldon Cotts, Vikas Berry, "Graphene Chemo-Phonics for Biosensor Applications: An Interfacial Raman Transducer", **Advanced Materials Interfaces**, 9, 30, 2200773, 2022
10. Roshan Nemade, Sheldon Cotts, Vikas Berry, "Plasma treatment for enhanced microbe-electrode interfaces: A bio-electronic sink", **Journal of Power Sources**, 544, 231834, 2022
11. Alireza Ahmadianyazdi, Ngoc Hoang Lan Nguyen, Jie Xu, Vikas Berry, "Glucose measurement via Raman spectroscopy of graphene: Principles and operation", **Nano Research**, 15, 10, 8697 2022
12. Ngoc Hoang Lan Nguyen, Sungjoon Kim, Garrett Kindemann and Vikas Berry, "COVID-19 Spike Protein Induced Phononic Modification in Antibody-Coupled Graphene for Viral Detection Application", **ACS Nano**, 15, 11743, 2021
13. Sanjay Behura*, Chen Wang, Yu Wen, and Vikas Berry*, "*Graphene/Semiconductor Heterojunction Sheds Light on Emerging Photovoltaics*", **Nature Photonics**, 13, 312–318, 2019
14. N. Mohanty, D. Moore, Z. Xu, T. S. Sreeprasad, A. Nagaraja, A. A. Rodriguez and V. Berry*, "*Nanotomy Based Production of Transferrable and Dispersible Graphene-Nanostructures of Controlled Shape and Size*", **Nature Communications**, 3, Article number: 844, 2012

PATENTS:

GRANTS

15. "Direct and sequential formation of monolayers of boron nitride and graphene on substrates", Michael R. Seacrist, Vikas Berry and Phong T. Nguyen, **US Patent US9029228 (2023)**
16. "Direct formation of graphene on semiconductor substrates and structures prepared thereby", M.R. Seacrist and Vikas Berry, **US Patent US20160233305 (2023)**
17. "Optical COVID-19 Detection System", Vikas Berry, Lan Nguyen and Sungjoon Kim, **US**

Patent 20220299506 (2022)

18. “*Production of Graphene Nanoribbons and Quantum Dots with Controlled Dimensions and Crystallographic Orientation*”, Vikas Berry, Nihar Mohanty and David Moore, **US Patent US9272911 (2016)**
19. “*Direct formation of graphene on semiconductor substrates*”, Michael R. Seacrist and Vikas Berry, **Patent Number US8884310 (2014)**
20. “*Fabrication of Ultra Long Necklace of Nanoparticles*”, Ravi Saraf, S. Niu, V. Berry, V. Maheshwari, **Patent Number: US7749561 (2010)**

9. Professional Development Activities

- Plenary and Keynote Lectures
 - Plenary: Graphene Conference (Rome, 2019), *Tsinghua University* (China 2011)
 - Keynote: TechConn (2020), RRR (2019), *ISFM* – (India 2018); *ICREA – Graphene Biosensors* (Spain - 2015); *IMST* (India, 2012)
- Founded two startups (Grapherry Inc., Viroscope Inc.) advancing graphene materials and airborne virus detection technologies.
- Mentored 3 postdocs, 18 PhD and 6 MS students; 8 former trainees now faculty members worldwide.
- Organizer of departmental workshops, and industrial advisory engagements.
- Implemented major curriculum innovations, including development of new BS concentrations, professional development modules, and practice-oriented graduate training pathways.
- Initiated and led research collaboration programs with Argonne National Laboratory, the University of Chicago, and multiple international academic partners.